Comparative Analysis of Custom Linear and Optimized Non-Linear Machine Learning Architectures for Post-Myocardial Infarction Complication Prediction

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It is very common that complications occur to a patient following a myocardial infarction (MI). Therefore, predicting them is crucial for patient care management. However, developing predictive models on clinical datasets which are moderately sized and highly dimensional presents an outstanding machine learning challenge.

This study aims to evaluate the capacity of models when dealing with strict tabular datasets. In order to do this, a Logistic Regression model was implemented from scratch by using gradient descent optimization with the goal of minimizing binary cross-entropy. Two advanced learning algorithms were compared with the performance of the baseline: a Multi-Layer Perceptron (MLP) and a Random Forest classifier. Lastly, due to the risk of overfitting given the limited datasize, methods such as early stopping and L2 regularization were included into the MLP pipeline.

An accuracy of 78.8% was achieved from the Logistic Regression implemented from scratch, this was indicated via a stratified 5-fold cross-validation. In contrast, the Random Forest ensemble captured feature interactions without clear signs of overfitting, achieving an accuracy of 82.0% and an Area Under the Curve (AUC) of 0.902. Overall, these results indicate that while custom linear baselines and regularized deep learning models perform similarly on strict tabular datasets, tree-based ensembles achieve better performance.

CCS Concepts: • **Computing methodologies** → **Machine learning**.

Additional Key Words and Phrases: Machine Learning, Myocardial Infarction, Random Forest, Logistic Regression

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1 Introduction

One of the leading causes of morbidity and mortality around the world is Myocardial Infarctions (MI), commonly known as heart attacks. Even though immediate survival rates for MI have improved with time due to big advancements in cardiovascular care, there is still a high risk for patients of developing serious complications when in the post-infarction period. These complications, such as arrhythmias and recurrent angina or cardiogenic shock and heart failure, have a big association with worse expectations for patients and high pressure on healthcare systems. Therefore, early prediction of these complications allows for more patient-specific treatment, which would improve outcomes from patients. In other words, the ability to identify in advance the patients who are at a high risk of these types of problems is very important in order to treat them correctly.

The increasing use of technology in electronic health records has made it possible to develop data-driven prognostic tools. Nevertheless, due to clinical dataset regulations such as patient privacy and medical data collection complexity, these datasets are "tabular" (structured in rows and columns) and have very high dimensionality

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but moderate size in patient records. Highly complicated algorithms, such as deep neural networks, often need very large quantities of data in order to generalize in an efficient manner. When these types of algorithms are used with moderately sized datasets, they also need strict generalization methods, such as early stopping and regularization, due to the high risk of overfitting. On the other hand, simpler algorithms don't have as high of a risk of overfitting but they may be less effective.

The overall focus of this study is the relationship between algorithmic complexity and data availability. While advanced architectures such as Multi-Layer Perceptrons and tree-based ensembles are normally said to be superior predictive tools, this superiority is challenged by simpler linear models when data is constrained, creating a need for empirical validation. Therefore, this paper investigates the following primary research question: When evaluated on a tabular clinical dataset utilizing a standardized stratified 5-fold cross-validation protocol, how does a custom-built Logistic Regression baseline compare against hyperparameter-optimized Random Forest and Neural Network architectures in predicting post-myocardial infarction complications across Accuracy, F1-Score, and Area Under the Curve (AUC) metrics?

In order to efficiently answer this question, this study establishes a foundational baseline by implementing a Logistic regression algorithm entirely from scratch, using gradient descent to minimize binary cross-entropy. The capacity of this baseline implemented from scratch will then be compared against two non-linear architectures: a Multi-Layer Perceptron (designed with strict early stopping methods to maximize generalization) and a Random Forest classifier. By using strict hyperparameter optimization and cross-validation for all these models, this comparative analysis aims to explain the balance between model complexity and predictive capacity for post-MI complication forecasting.

2 Related Work

Extensive research has been conducted on the application of machine learning techniques to myocardial infarction and other cardiovascular conditions. Across the literature, datasets typically range from approximately 300 to 2500 instances and contain between 13 and 75 features, reflecting the structured and moderately sized nature of clinical tabular data. These characteristics strongly influence model selection and performance.

Several studies have explored the comparative performance of multiple machine learning algorithms for heart disease prediction. Igole et al. [4] evaluated seven different models and reported highly competitive results, achieving an accuracy of 91.51% using a Support Vector Machine, with Logistic Regression closely following at 91.44%. These findings highlight that both linear and nonlinear models can perform well on structured medical datasets. Similarly, Sooriyaperakasam et al. [8] compared Random Forest and Logistic Regression on a dataset of 303 samples with 14 features, achieving an accuracy of 86%, while Ghosh et al. [2] reported an overall accuracy of 84%, with Random Forest slightly outperforming other models at 88%.

Other studies have focused specifically on myocardial infarction outcomes. Zhang et al. [10] analyzed data from 1,362 patients to predict major adverse cardiovascular events and found that an Artificial Neural Network achieved the highest performance (AUC = 80.4%), outperforming Random Forest (72.67%) and Logistic Regression (71.77%). These results suggest that more complex models may capture nonlinear relationships more effectively when sufficient data is available. However, performance differences across studies indicate that no single model consistently outperforms others in all settings.

The continued relevance of Logistic Regression in clinical prediction is well established. A comprehensive review by Hua et al. [3], analyzing 41 peer-reviewed studies, concluded that Logistic Regression remains a cornerstone in clinical risk prediction due to its interpretability and robust handling of binary outcomes. Its transparency makes it particularly valuable in healthcare, where understanding the relationship between predictors and outcomes is essential.

Despite promising results, model performance is highly dependent on dataset characteristics. Li et al. [5] investigated the impact of regularization parameters and found only marginal performance differences across varying settings, with accuracies ranging from 83% to 86%, suggesting limited sensitivity to parameter tuning in certain contexts. In contrast, Fourkiotis et al. [1] demonstrated the impact of distributional shifts by training models on male patients and evaluating them on female patients, achieving 85% accuracy but highlighting challenges in generalization across populations. Furthermore, Obasi et al. [7] reported significantly lower performance (59.7% accuracy) despite a relatively large dataset of 1,990 observations, emphasizing that dataset quality and feature relevance are as important as dataset size.

Dataset size remains a critical factor in model reliability. Zantvoort et al. [9] showed that small datasets ($N \leq 300$) often lead to overestimated predictive performance due to overfitting, particularly in complex models. Their findings indicate that while datasets of around 500 samples can reduce overfitting, stable performance generally requires between 750 and 1500 samples. Additionally, Naser [6] highlights that methods for ensuring generalization in small or sparse datasets remain underexplored.

Overall, existing research demonstrates that while advanced models can offer improved predictive performance, their effectiveness depends heavily on dataset size, quality, and distribution. Consequently, systematic comparisons between linear and nonlinear models remain essential for identifying the most suitable approach for clinical prediction tasks.

3 Methodology

In order to answer the research question, the experimental methodology has been structured into three phases: Data Preprocessing and Target Formulation (rigorous data preprocessing), Custom Linear Baseline Formulation (mathematical derivation of a custom baseline model), and Advanced Non-Linear Architectures (the deployment of advanced non-linear architectures). All of the models were evaluated under identical constraints, which was ensured through the standardization of the preprocessing pipeline.

3.1 Data Preprocessing and Target Formulation

The initial dataset contained 1,700 patient records which contained various clinical features after a myocardial infarction. Achieving a definitive predictive target was the primary action to be done. The raw data contained several individual variables which represented post-MI complications. These variables were combined into a unified binary target in order to create a binary prediction model. The unified binary target was: patient records exhibiting one or more complications in the positive class, and patient records with nothing in the negative class. Lastly, all of the original features related to complications were removed from the feature space in order to prevent data from leaking.

In addition, too much imputation may have resulted in an increase in bias, therefore, a threshold was set: features with more than 30% missing values were removed. This helped to keep useful clinical data against the errors created by too much imputation, resulting in 105 features. The missing values for the remaining features were filled using median imputation, which is very strict against outliers, while preserving data distribution accordingly.

3.2 Custom Linear Baseline Formulation

A simple linear classifier was derived from scratch in order to establish a performance baseline. Specifically, a Logistic Regression algorithm modeled the probability of a patient in developing post-MI complications. The model uses the 105 features in order to calculate a weighted linear combination, converted into a probability between zero and one through a sigmoid activation function.

Also, the use of a batch gradient descent algorithm helped achieve optimization. During the training, the model iteratively updated the weights from its features and the bias with the goal of minimizing binary cross-entropy (log-loss) cost function. By computing the partial derivatives of the cost function and moving in the opposite direction of the steepest ascent, which was scaled by the learning rate, the algorithm descended towards the global minimum. Convergence was shown by monitoring the loss across 2,000 iterations, which resulted in an interpretable baseline implemented from scratch.

3.3 Advanced Non-Linear Architectures

To evaluate if increased model complexity leads to better performance, the baseline was compared against two advanced non-linear architectures.

Firstly, a Multi-Layer Perceptron (MLP) used Rectified Linear Unit (ReLU) activation functions in order to approximate complex patterns. A sigmoid function was used in the final output layer, and the Adam solver was used in order to optimize the network. Also, strict generalization methods such as early stopping and L2 regularization were integrated in order to minimize the risk of overfitting. These methods' biggest impact was that they made the network identify patterns rather than memorize them during training, proving their functionality.

Following, a Random Forest Classifier. It is an ensemble learning method, which constructs independent decision trees during training. Bootstrap aggregating (bagging) uses randomness and evaluates only a random subset of features at each node split, which helps reduce their correlations. It aggregates individual tree outputs through majority voting. This architecture models non-linear relationships between features without explicit feature engineering. The ensemble aggregation helps lower individual tree variance, making the Random Forest's risk of overfitting lower. Both models were optimized through hyperparameter tuning with stratified cross-validation.

4 Experimental Setup

A standardized experimental and validation protocol was set in order to evaluate the predictive performance of the baseline against the advanced models. This section outlines: Dataset Configuration and Stratification, Cross-Validation Protocol and Evaluation Metrics.

4.1 Dataset Configuration and Stratification

The finalized dataset contained 1,700 patient records and 105 features. The target variable, which represented the presence or absence of post-MI complications, showed a moderately sized class imbalance, where approximately 61% of the cases were in the positive class(presence) and the other 39% were in the negative class(absence). The dataset was divided into a stratified 80/20 split in order to ensure that the models were trained and evaluated on representative samples of the full dataset. Specifically, 80% of the data was used for model training and hyperparameter optimization, while the other 20% was left to be unseen test set for final performance validation. This mechanism ensured that the original distribution was preserved during the training and testing sets.

4.2 Cross-Validation Protocol

In order to prevent overfitting and ensure robust model generalization, a k-fold cross-validation strategy was used in the training phase for all of the models. Specifically, a stratified 5-fold cross-validation protocol was used. During this process, the training data was divided into five partitions of the same size. Each model was trained iteratively on four of the partitions and then tested on the last one left. This procedure was repeated five times, switching the partition used in the testing part in each iteration. By averaging the performance across all five iterations, the variance of the models was calculated.

4.3 Evaluation Metrics

Given that the context of this study is clinical and that there exists a class imbalance in the dataset, relying on accuracy alone for evaluation was not sufficient. Therefore, three metrics have been used to evaluate model performance:

- **Accuracy:** Used as a baseline measure to represent the overall proportion of correct predictions across both of the complication classes (positive and negative).
- **F1-Score:** It is the mean of precision and recall. It was prioritized during the hyperparameter tuning because it heavily penalizes errors in predictions, ensuring that the model does not misclassify individuals and identifies correctly when they are high-risk patients.
- **Area Under the Curve (AUC):** Derived from the Receiver Operating Characteristic (ROC) curve, it improves the model's ability to differentiate between patients who developed complications and those who didn't.

5 Results

The predictive performance of the baseline and the advanced models was evaluated by using the 5-fold cross-validation protocol and the hold-out(80/20) test set.

5.1 Baseline Mathematical Convergence

Before comparing the models, the convergence of the Logistic Regression baseline was evaluated. By tracking the binary cross-entropy during the training phase, the model showed a clear convergence pattern. As shown in Figure 1, the cost function decreased from its initial weights and stabilized near a minimum cost of 0.39 within 2,000 iterations, indicating that the model learned appropriate weights.

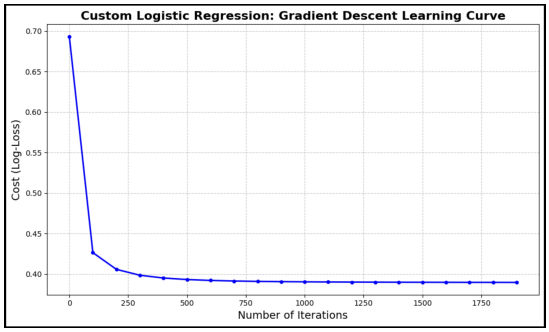


Fig. 1. The learning curve of the custom Logistic Regression model, demonstrating the minimization of the log-loss cost function over 2,000 iterations.

5.2 Comparative Model Performance

Before evaluating the models on the unseen test set, the Logistic Regression baseline achieved a predictive accuracy of 78.82%. Also, the Multi-Layer Perceptron (MLP) stabilized at a very near accuracy of 79.0%. On the other hand, the Random Forest ensemble showed a better performance, achieving the highest accuracy at 82.0%.

In order to ensure these point-estimate accuracies were not anomalies, the models were compared using their mean F1-Scores obtained from the stratified 5-fold cross-validation process (Figure 2a). The Logistic Regression and the MLP showed F1-Scores of 0.815 and 0.819, respectively. The Random Forest showed a higher score,

obtaining an F1-Score of 0.864. In addition, the standard deviation error bars indicated that the Random Forest maintained the most stability across all of the data folds.

Finally, all of this was confirmed by the Receiver Operating Characteristic (ROC) evaluation (Figure 2b). The Random Forest achieved a superior Area Under the Curve (AUC) of 0.902, showing higher ability across all classification thresholds, compared to the MLP (AUC = 0.863) and the Logistic Regression (AUC = 0.854).

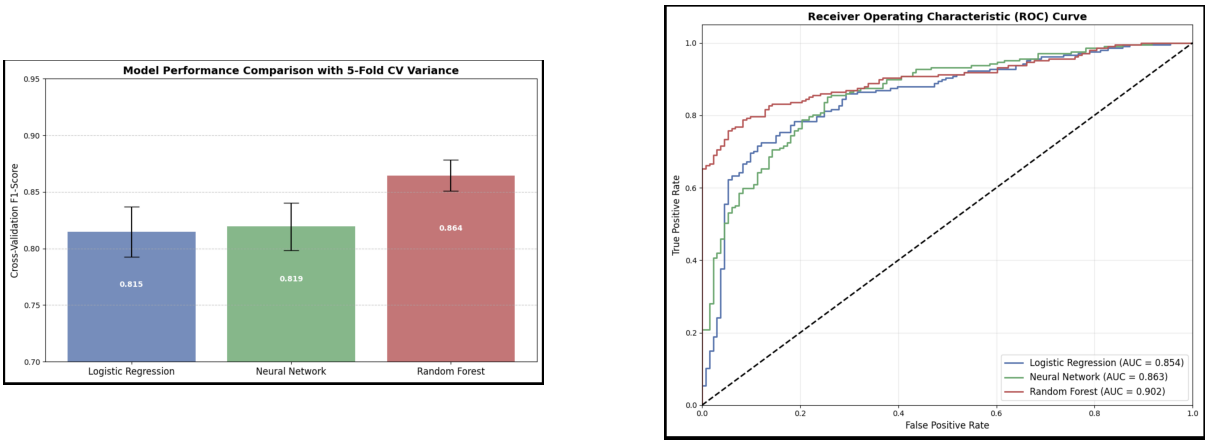


Fig. 2. (a) Mean 5-Fold Cross-Validation F1-Scores with standard deviation variance. (b) Receiver Operating Characteristic (ROC) curves depicting the discriminative capability of the three optimized architectures.

5.3 Feature Importance

As an ensemble of decision trees, the Random Forest algorithm provides a measure of the importance of each feature. The model identified several important variables that were important for predicting the identification of high-risk patients. Figure 3 shows the relative importance of the top ten clinical features based on the ensemble's Gini impurity metrics, with Age and Sex being the most important features for predicting post-MI complications.

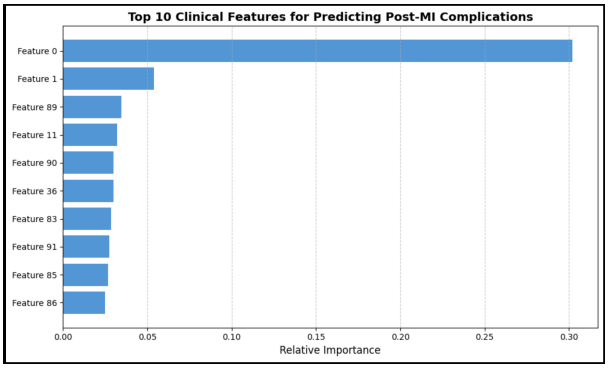


Fig. 3. Relative feature importance of the top ten clinical predictors as determined by the optimized Random Forest ensemble.

6 Discussion and Future Work

The results of this comparative analysis show the impact that dataset characteristics and volume have on the performance of machine learning algorithms in clinical settings. The main goal of this study was to evaluate a linear model implemented from scratch against advanced non-linear models under limitations of a moderately sized tabular dataset.

6.1 Algorithmic Complexity vs. Data Volume Constraints

A very remarkable finding of this study is the near-identical performance between the Logistic Regression baseline (78.8% accuracy) and the Multi-Layer Perceptron (79.0% accuracy). In theoretical machine learning, deep neural networks can approximate complex functions very efficiently, allowing them to model complex , non-linear patterns in the data. However, these architectures require large amounts of training data in order to learn useful feature representations effectively.

When applied to the limited volume of the experimental dataset (1,700 records), the MLP had a high risk of overfitting. In order to address this, regularization methods , such as early stopping mechanisms or L2 regularization penalties, were applied during the training phase. While they successfully prevented the MLP from overfitting the training data, they also constrained the capacity of the model. As a result of this, the MLP behaved similarly to a model such as a Logistic Regression, resulting in an Area Under the Curve (AUC = 0.863) that showed very little improvement over the foundational Logistic Regression model (AUC = 0.854). These results indicate that for moderately sized datasets, using deep learning models does not provide a clear advantage over linear models such as Logistic Regression.

6.2 The Superiority of Tree-Based Ensembles on Tabular Data

In contrast to the baseline and the MLP, the Random Forest ensemble showed a clear performance advantage across all metrics, achieving an F1-Score of 0.864 and an AUC of 0.902. The better performance of this architecture can be explained by its structure characteristics. Clinical biometrics often show non-linear interactions; for instance, the combined presence of two specific physiological markers may indicate a much higher risk of post-MI complications than considering each marker separately. Far from linear boundaries or regularized network weights, the decision trees within the Random Forest are able to capture these non-linear feature interactions through splitting the data at each node. Furthermore, the ensemble's use of bootstrap aggregating (bagging) and random feature selection at each node split helped reduce the variance of the individual trees. This allowed the Random Forest to model complex relationships deeply while avoiding the overfitting issues observed in the MLP.

6.3 Clinical Interpretability

In addition to predictive performance, the Random Forest was very interpretable. The algorithm's Gini impurity metrics showed Age and Sex as the most important predictors for post-MI complications. In practice, models that offer clear feature importance allow medical practitioners to understand the reason behind a prediction, increasing trust in the model.

6.4 Future Work

While this study suggests that tree-based ensembles perform better for moderately sized tabular datasets, future research should explore the point at which deep learning performs better than ensemble methods. We expect that if the clinical cohort were increased, exceeding 50,000 patient records, the larger dataset would allow neural networks to be less affected by regularization and learn more complex patterns in the data. Additionally, future work could use unstructured medical data, such as post-MI echocardiogram imaging or electrocardiogram time-series data, where deep learning models perform well.

7 Conclusion

Being able to predict complications following a myocardial infarction is a very important step in improving patient outcomes. This study suggests that while deep learning models perform well in many predictive tasks, their application to moderately sized, tabular datasets is limited by the size of the dataset. A Logistic Regression model, implemented entirely from scratch, performed competitively with a regularized Multi-Layer Perceptron. Overall, the Random Forest ensemble was the best performing model, being able to capture non-linear clinical interactions without overfitting. This research shows that when dealing with moderately sized clinical datasets, tree-based ensembles provide a strong balance of predictive performance and interpretability.

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Information Sheet

Like the assignment rubric requirements say, this section details the division between the algorithmic components implemented entirely from scratch, those utilizing standard machine learning libraries, and the assistive tools used during the project.

1. Components Built from Scratch

The baseline model was developed without the use of high-level predictive or optimization libraries. Relying exclusively on **NumPy** for matrix operations and mathematical calculations, the following components were programmed entirely from scratch:

- **Logistic Regression Algorithm:** The core predictive logic calculating the dot product of the feature space and weight matrix.
- **Sigmoid Activation Function:** The non-linear mapping function to bound continuous outputs into a valid probability space.
- **Binary Cross-Entropy (Log-Loss):** The mathematical cost function manually coded to evaluate prediction error during training.

- **Batch Gradient Descent Optimizer:** The iterative optimization algorithm, including the manual calculation of partial derivatives, gradient updates, and learning rate application to achieve convergence over 2,000 iterations.

2. Components Utilizing External Libraries

To handle data manipulation, advanced non-linear architectures, and standardized evaluation, the following Python libraries were utilized:

- **Pandas:** Used for initial tabular data loading, feature manipulation, and the structural mathematical aggregation of the granular post-MI complications into the unified binary target variable.
- **Scikit-Learn (Data Preprocessing):** Utilized for resolving missing values via the `SimpleImputer` (median strategy) and partitioning the dataset using `train_test_split` and `StratifiedKFold` for cross-validation.
- **Scikit-Learn (Advanced Architectures):** Utilized for the implementation, hyperparameter tuning, and execution of the advanced non-linear models: `MLPClassifier` (Multi-Layer Perceptron) and `RandomForestClassifier`.
- **Scikit-Learn (Evaluation Metrics):** Utilized to compute standardized performance metrics, specifically `accuracy_score`, `f1_score`, and `roc_auc_score`, as well as for generating the data points required for the ROC curve visualizations.

3. Use of Generative AI Tools

In accordance with academic transparency and integrity guidelines, we disclose the use of Large Language Models (specifically Google's Gemini) as an assistive tool during the development of this project. Gemini was exclusively utilized for the following supportive tasks:

- **Ideation:** Initial brainstorming and conceptual structuring during the early phases of the project.
- **Debugging:** Troubleshooting and resolving specific technical programming errors encountered during the model implementation phase.
- **Document Formatting:** Providing guidance on LaTeX syntax and structural formatting to facilitate the transition of the final manuscript from a standard word processor into the Overleaf environment.
- **Language Assistance:** Minor support in refining phrasing and vocabulary for clarity and readability. All technical content, methodology, and conclusions were developed independently by the authors.